MATH 14–Skeletal Notes

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How to Use These Notes

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

Wherever a derivation, a solution, or a sketch belongs, there is a blank box, and in front of it a short parenthetical instruction saying what has to be done there — which formula to reach for, what to watch out for, what to conclude — but not how to do it. The figures, the animations, and the interactive applets that follow a box are the check on what you wrote in it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the "PDF" button in the navigation bar gives you the whole book as one file to print from. That file is rebuilt every time new notes are posted, so download a fresh copy rather than printing from an old one. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

The assignments, with their answer boxes and the AI tutor, and the review problem sets live online only, in the chapter *Assignments and Review Problems* of the web version of this book at mahmadi-ops.github.io/MATH14-Skeletal; they are meant to be worked there, not on paper, so they are not in the PDF.

The big picture

The map below gathers every integral of this book on one page. Parametrization feeds the line integrals; the five theorems act as bridges, trading an integral over a boundary for an integral over the region, surface, or solid it bounds; and each multiple integral carries its coordinate substitutions. An arrow between two topics means the second builds on the first, and the gold arrows mark the theorem and substitution bridges between the integral types. Every topic on the map is a link to the part of the book where it is covered. For comfortable reading, open the map full size in a new tab.Interactive map of the book's integrals (view in the HTML version).

Contents

Chapter 1

Line Integrals

1.1 Parametrization of Curves

Objectives

- Describe the motion of a point in the plane with a pair of parametric equations $x = f(t)$ and $y = g(t)$, and understand the parameter t as recording *when* the point visits each place on the curve.
- Sketch a parametric curve from a table of values, and indicate the direction of increasing t in which the curve is traced.
- Find parametric equations for common curves, such as lines, circles, and ellipses, and eliminate the parameter to recover a Cartesian equation when possible.
- Distinguish the **intersection points** of two parametric curves from their **collision points**, where two moving objects reach the same place at the same time.
- Write a space curve as a vector function $\mathbf{r}(t)$, and parametrize the curve in which two surfaces meet.

Imagine that you would like to describe the motion of an object whose motion is confined to a plane. If you know how its Cartesian coordinates, x and y, change as time passes, you will be able to draw a curve that represents the trajectory of its motion. Mathematically, we will need two equations to demonstrate how the $x-$ and y-coordinates evolve in time t, which are known as the parametric equations. This section reviews that language and then extends it to curves in space, where it becomes the main tool for everything that follows.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1.1 Parametric Equations

Definition 1.1.1 Parametric Equations. If x and y are given as functions of a third variable t, called a **parameter**, by

$$x = f(t), \qquad y = g(t), \tag{1.1.1}$$

then the points $(x, y) = (f(t),\, g(t))$ trace out a curve in the plane, called a **parametric curve**. The equations $x = f(t)$ and $y = g(t)$ are the **parametric equations** of the curve. ◊

As t runs over its interval, the point $(f(t), g(t))$ moves, and the trail it leaves behind is the curve. The parameter therefore carries more information than the picture does: it also records *when* the moving point is at each place, as in Figure 1.1.2.

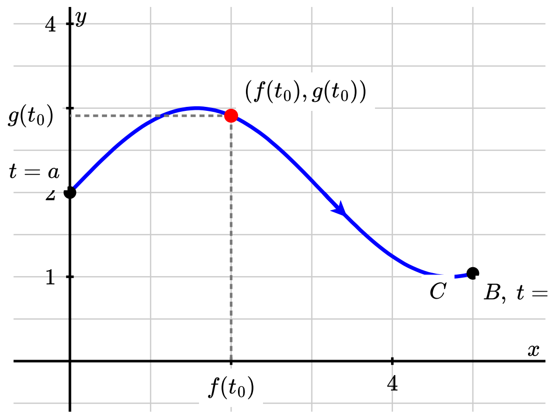
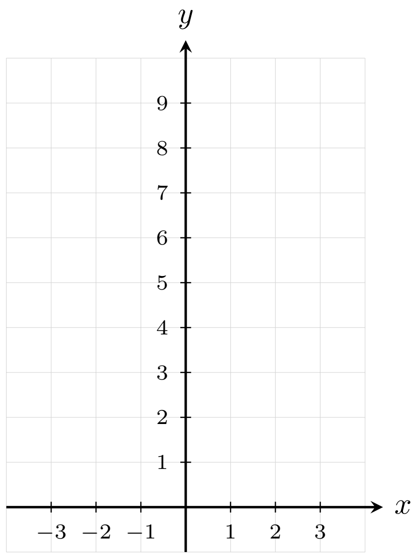


Figure 1.1.2 The trajectory of a moving point whose coordinates are given by the parametric equations $x = f(t)$, $y = g(t)$. At the instant t_0 the point is at $(f(t_0), g(t_0))$.

As an example, consider $x = t$ and $y = t^2$. Fill in Table 1.1.3, plot the seven points on the grid below, and join them. (Then eliminate the parameter to name the curve, and mark on your sketch the direction in which it is traced as t increases.)

Table 1.1.3 Points on the curve $x = t$, $y = t^2$

t	x	y
-3		
-2		
-1		
0		
1		
2		
3		



Eliminating t:

Name of the curve, and its direction:

As our second example, consider the equation of the circle centered at $(0, 0)$ with unit radius,

$$x^2 + y^2 = 1. \tag{1.1.2}$$

Here, we would like to parametrize x and y in terms of a single parameter t, as in (1.1.1), in such a way that by varying the value of t we can reconstruct the circle (1.1.2). Find such a pair.

the parametrization:

interval of t, and the check:

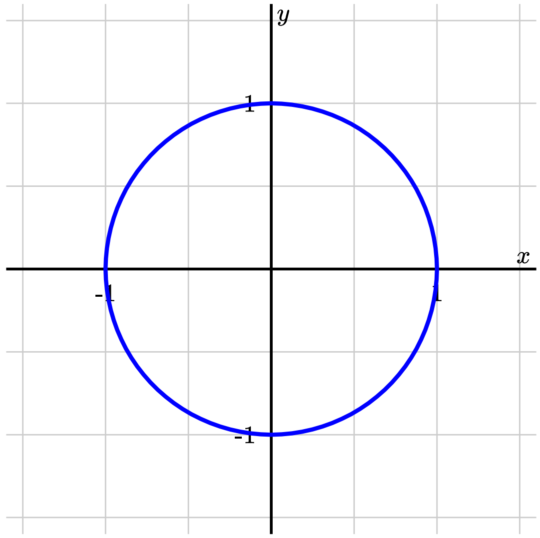


Figure 1.1.4 The unit circle $x^2 + y^2 = 1$.

Checkpoint 1.1.5 Question. Can you think of another way to write a parametric form of the circle $x^2 + y^2 = 1$?

workspace

1.1.2 Some Examples of Parametrizing Curves

Example 1.1.6 Parametrizing Curves. Find the parametric equations corresponding to the following curves.

1. The line segment connecting the two points $(1, 0)$ and $(0, 1)$.
2. The ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$.
3. The circle of radius 2 centered at $(2, 0)$.

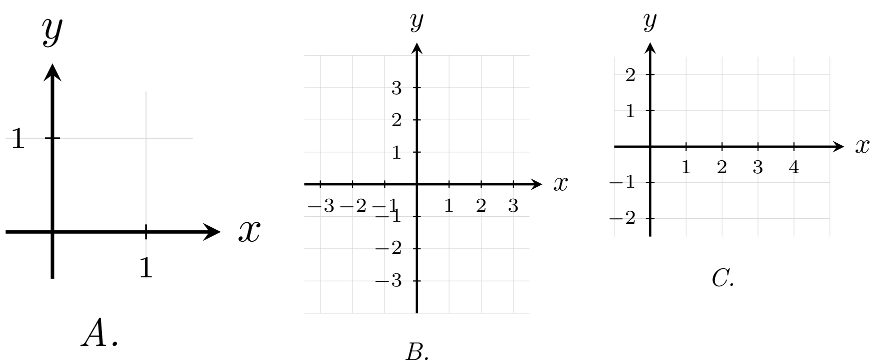
Solution.

A.

B.

C.

Now sketch the three curves on the grids below, marking the point reached at $t = 0$ and the direction of increasing t on each.



□

1.1.3 Intersection Points versus Collision Points

A parametrization carries more information than the curve it traces: it also records *when* the moving point visits each place on the curve. Because of this, there are two different questions we can ask about two parametric curves

$$\mathbf{r}_1(t) = \langle f_1(t),\, g_1(t)\rangle\,,$$
$$\mathbf{r}_2(t) = \langle f_2(t),\, g_2(t)\rangle\,,$$

thought of as the trajectories of two particles.

Definition 1.1.7 Intersection Points and Collision Points. A point P is an **intersection point** of the two curves if P lies on both curves; that is, if there are parameter values t and s, not necessarily equal, with

$$\mathbf{r}_1(t) = \overrightarrow{OP} = \mathbf{r}_2(s).$$

A point P is a **collision point** of the two particles if both particles are at P at the same time; that is, if there is a single value of t with

$$\mathbf{r}_1(t) = \mathbf{r}_2(t) = \overrightarrow{OP}.$$

◊

An intersection point is a statement about the two *paths*: they cross, like two roads on a map. A collision point is a statement about the two *motions*: the cars are at the crossroads at the same instant. Every collision point is automatically an intersection point, but, as the next two examples show, an intersection point need not be a collision point.

In the next two examples the two paths are always the same: the parabola $y = x^2$ and the line $y = x + 2$, which meet at the two points

$$(-1, 1) \qquad \text{and} \qquad (2, 4). \tag{1.1.3}$$

Only the schedule of the second particle will change.

Example 1.1.8 The paths cross, but the particles never meet. Two particles move in the plane with position vectors

$$\begin{aligned} \mathbf{r}_1(t) &= \left\langle t,\ t^2 \right\rangle, \\ \mathbf{r}_2(t) &= \left\langle -t,\ 2 - t \right\rangle, \qquad t \in \mathbb{R}. \end{aligned}$$

Find all intersection points of their paths, and all collision points of the particles.

Solution.

the two paths:

collision equations:

when each particle reaches each crossing:

□

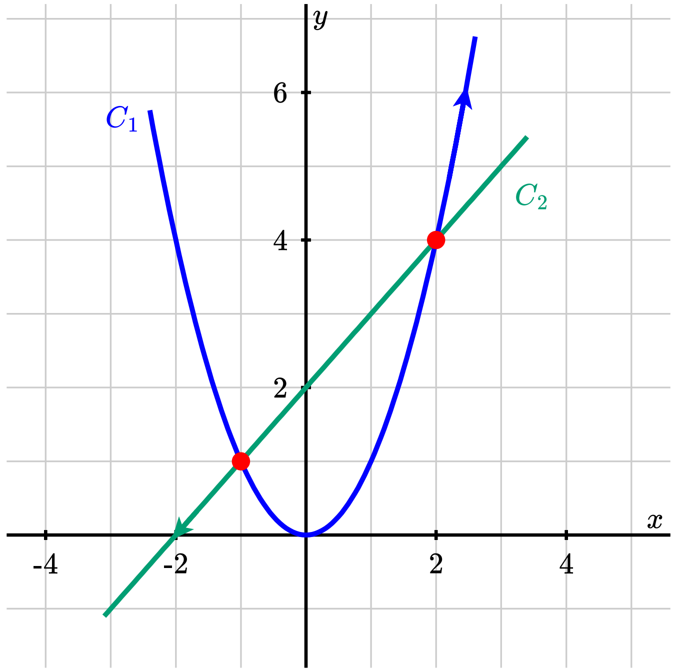


Figure 1.1.9 The parabola C_1 and the line C_2; compare with what you found.

Example 1.1.10 The same paths, a different schedule—now they collide. Keep the first particle, but re-schedule the second one:

$$\mathbf{r}_1(t) = \left\langle t,\ t^2 \right\rangle,$$
$$\mathbf{r}_2(t) = \langle 2t-2,\ 2t \rangle, \qquad t \in \mathbb{R}.$$

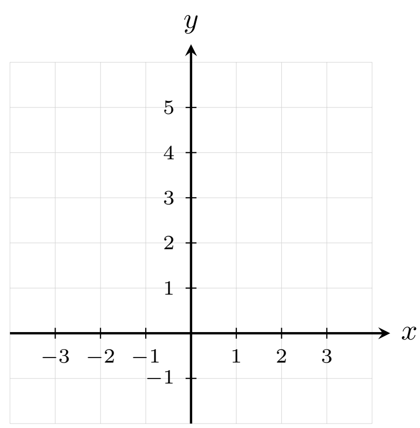
Find the intersection points of the paths and the collision points of the particles.

Solution.

the paths:

collision equations:

the other crossing:



□

Checkpoint 1.1.11 Keep $\mathbf{r}_1(t) = \left\langle t, t^2 \right\rangle$. Find a parametrization $\mathbf{r}_2$ of the line $y = x + 2$ for which the two particles collide at *both* intersection points.

$\mathbf{r}_2(t) =$

the collision equations it produces:

1.1.4 Curves in Space and Vector Functions

Assume that we are interested in describing the path of an object in space. To this aim, we can specify its location at each instant of time; that is, we can write its coordinates in $\mathbb{R}^3$ as

$$x = f(t), \qquad y = g(t), \qquad z = h(t). \tag{1.1.4}$$

Here t is the time parameter, which varies from some initial time t_i to some final time t_f, that is, $t \in [t_i, t_f]$. Nothing has changed except the bookkeeping: where a plane curve needed the two equations (1.1.1), a space curve needs three.

There is a second, more compact way to look at (1.1.4), and it is the point of view we will use for the rest of the course. Instead of three separate coordinate equations, collect the coordinates into a single **vector function** — a function whose input is a real number and whose output is a vector:

$$\begin{aligned}\mathbf{r}(t) &= \langle f(t),\ g(t),\ h(t)\rangle \\ &= f(t)\,\mathbf{i} + g(t)\,\mathbf{j} + h(t)\,\mathbf{k}, \qquad t \in [t_i, t_f].\end{aligned} \tag{1.1.5}$$

Think of $\mathbf{r}(t)$ as the **position vector** of the moving object: an arrow drawn from the origin to the object's location at time t. As t increases, the arrow swings around and changes length, and its *tip* sweeps out the curve. So a parametrization can be read in two equivalent ways.

- **Coordinatewise.** Three ordinary scalar functions f, g, h of one variable, one for each coordinate, as in (1.1.4).
- **Vectorially.** One vector-valued function $\mathbf{r}$ of one variable, as in (1.1.5). The curve is the set of tips of the vectors $\mathbf{r}(t)$.

The two descriptions carry exactly the same information, but the vector form is the more useful one: it lets us differentiate and integrate a whole motion at once, and $\mathbf{r}'(t)$ will turn out to be the velocity of the moving object. The same device works in the plane, where $\mathbf{r}(t) = \langle f(t), g(t)\rangle$; the position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ of Example 1.1.8 were already written this way.

Example 1.1.12 A helix. Consider the object whose coordinates are parametrized by

$$x = \cos(2t), \qquad y = \sin(2t),$$

$$z = t, \qquad t \in [0, \pi]. \tag{1.1.6}$$

Write the motion as a vector function and describe the path.
Solution.

$\mathbf{r}(t) =$

the shadow on the xy-plane, and how often it goes around:

what z does:

the path, its start, and its end:

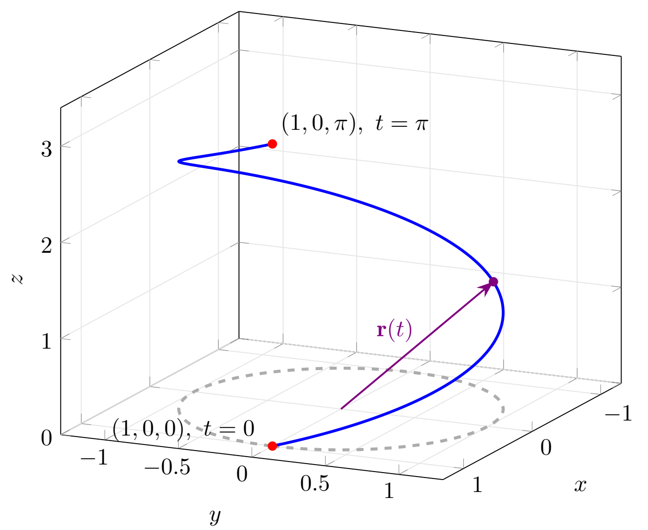


Figure 1.1.13 One turn of the helix $\mathbf{r}(t) = \langle \cos(2t), \sin(2t), t \rangle$, $0 \le t \le \pi$. The violet arrow is the position vector $\mathbf{r}(t)$ at $t = 3\pi/8$; its tip rides along the curve as t increases. The dashed gray circle is the shadow of the path on the plane $z = 0$.

□

There are situations in which a curve in space arises as the intersection of two surfaces. In the next two examples we turn such a description into a vector function $\mathbf{r}(t)$.

Example 1.1.14 Where a plane cuts a cylinder. Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

Solution.

$x(t)$ and $y(t)$:

$z(t)$:

$\mathbf{r}(t) =$

check at $t = 0,\ \pi/2,\ \pi,\ 3\pi/2$:

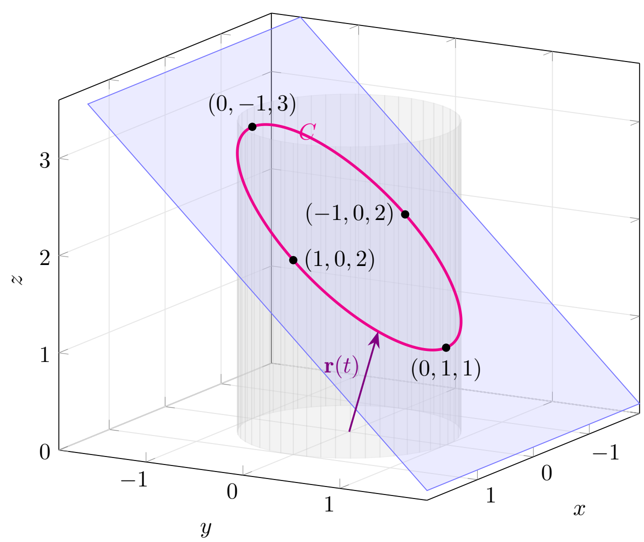


Figure 1.1.15 The cylinder $x^2 + y^2 = 1$, the plane $y + z = 2$, and the ellipse C in which they meet. The violet arrow is the position vector $\mathbf{r}(t)$ of your parametrization at $t = \pi/4$.

□

Example 1.1.16 Where two curved surfaces meet. Parametrize the intersection of the surfaces

$$z = \sqrt{1 + x^2 + y^2} \qquad \text{and} \qquad z = \sqrt{4 - 4x^2 - y^2}.$$

Solution.

$x(t)$ and $y(t)$:

$z(t)$, and $\mathbf{r}(t) =$

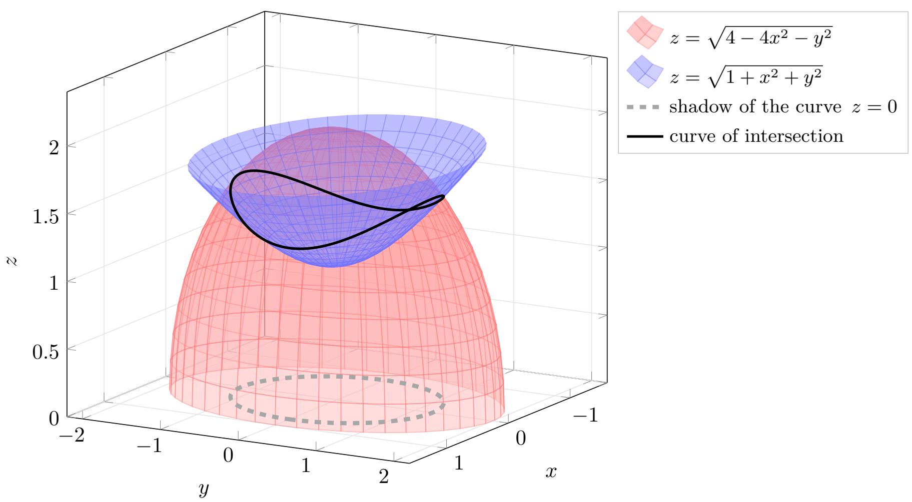


Figure 1.1.17 The upper sheet of $z = \sqrt{1 + x^2 + y^2}$ (blue) and the dome $z = \sqrt{4 - 4x^2 - y^2}$ (red) meet along the closed curve drawn in black. Its shadow in the plane $z = 0$ is the dashed ellipse; compare it with the equation you found.

□